

1. Alice can utilize a technique called superdense coding. to transmit two bits of information ("10" in this case)

1. Entangled state: Alice performs specific operations on her qubit depending on the two bits she wants to send:

- if first bit is 0, Alice keeps her qubit as is.
- if the first bit is 1, Alice applies a flip operation (NOT gate) to her qubit.

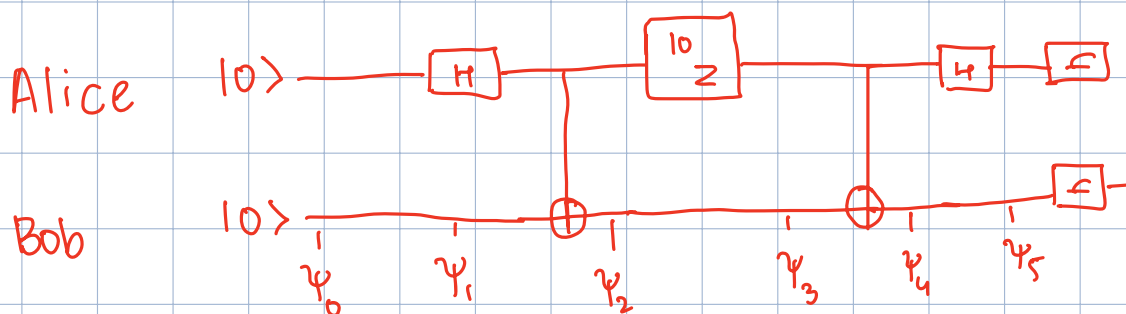
Then, depending on the second qubit

- if the second bit is 0, Alice does nothing further.
- if second bit is 1, Alice applies a phase shift operation.

Therefore, the operations Alice performs on her qubit are:

- To send "00": $I \otimes I$
- To send "01": $I \otimes X$
- To send "10": $Z \otimes I$
- To send "11": $X \otimes Z$

Alice transmits her qubit to Bob through the quantum channel.



$$\psi_0 = |00\rangle$$

$$\psi_1 = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

$$\psi_2 = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

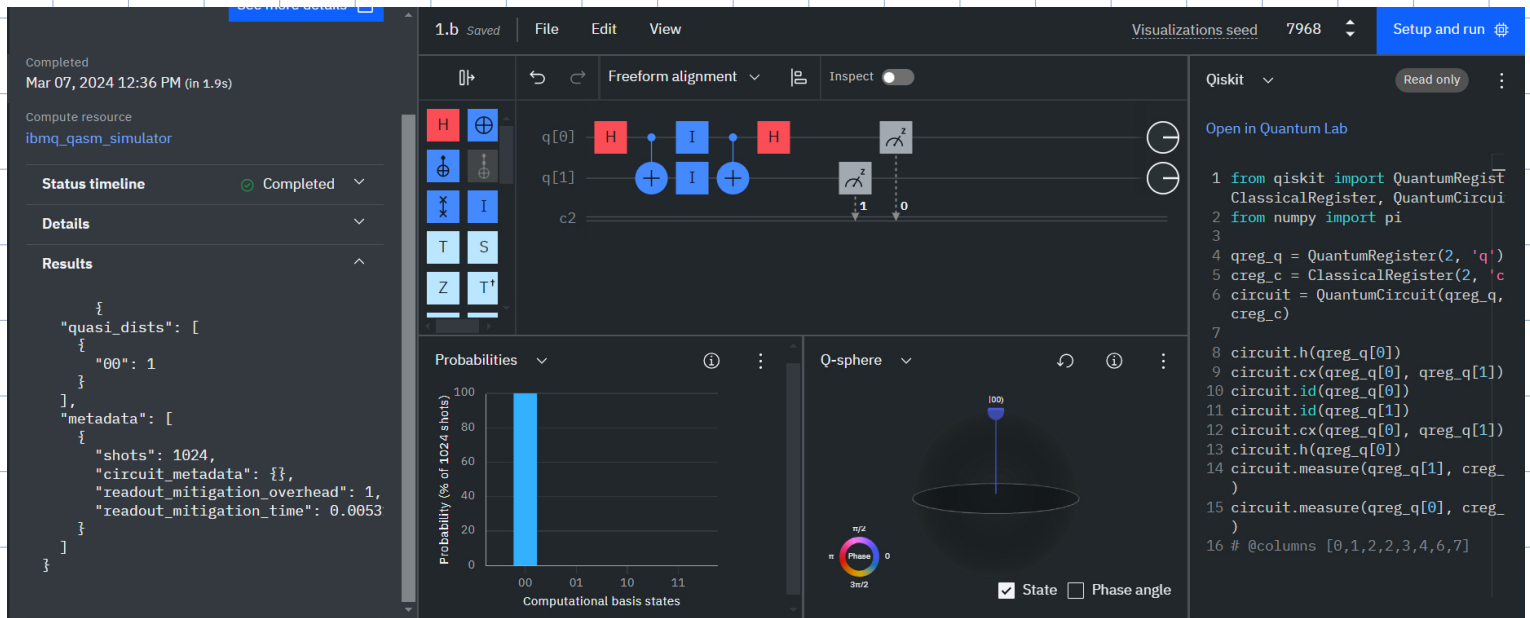
$$\psi_3 = \frac{1}{\sqrt{2}} (|00\rangle - |11\rangle)$$

$$\psi_4 = \frac{1}{\sqrt{2}} (|00\rangle - |10\rangle)$$

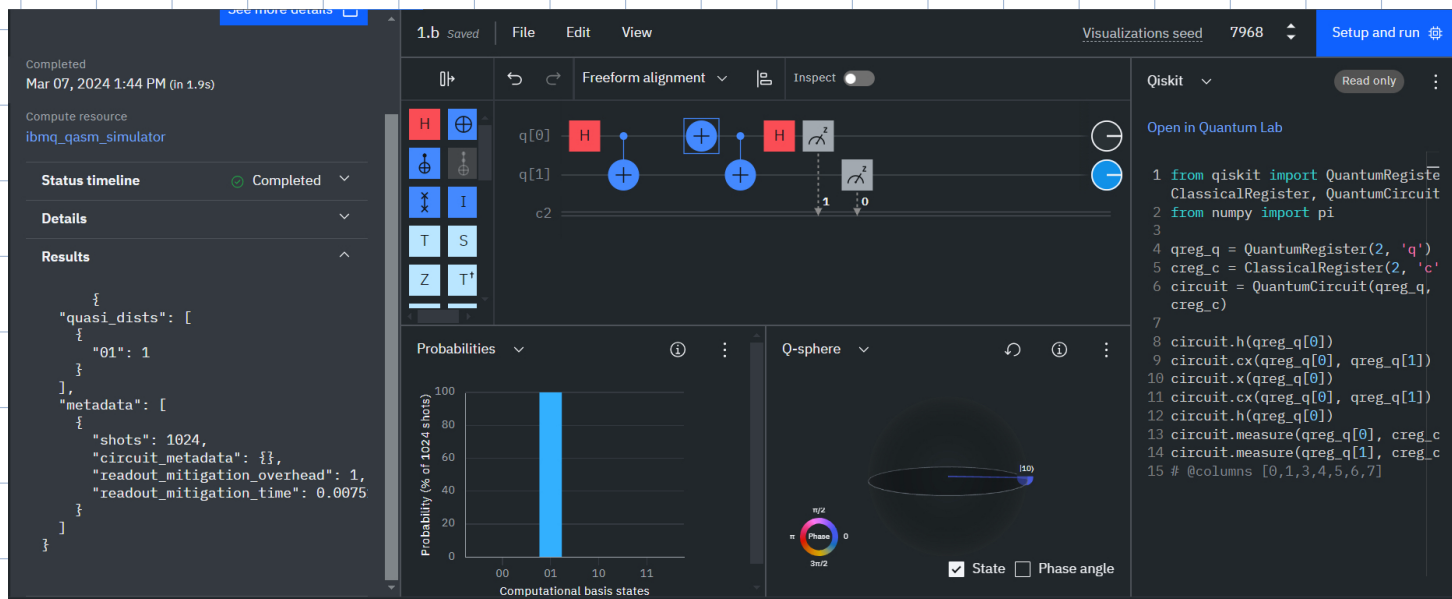
$$= \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes |0\rangle$$

$$\psi_5 = |1\rangle \otimes |0\rangle$$

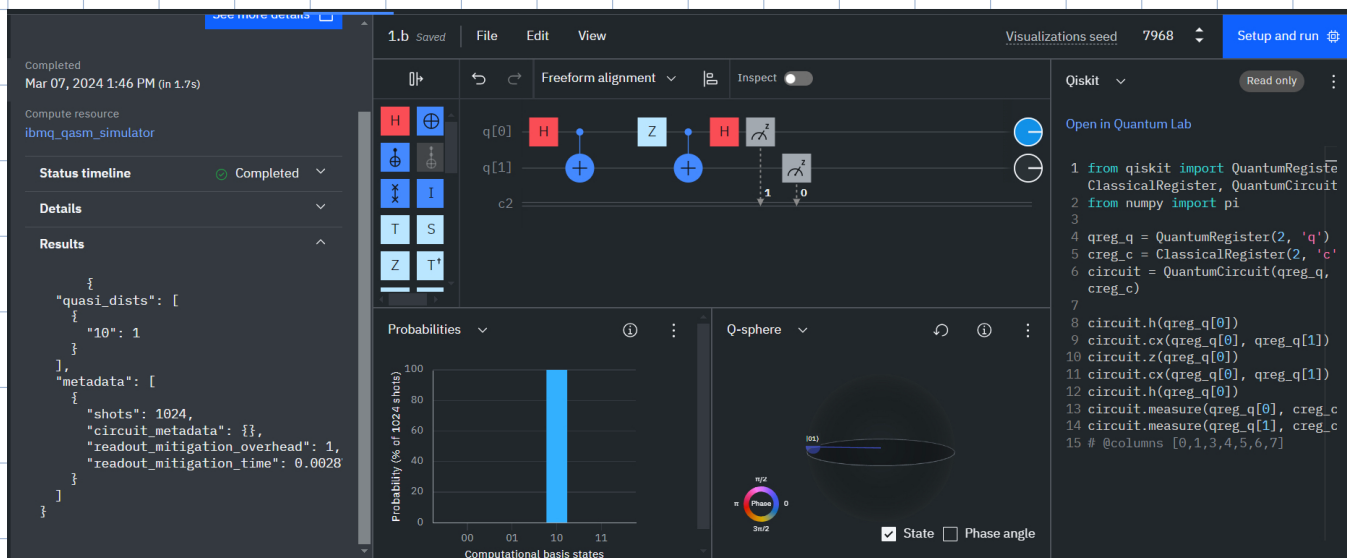
2. Alice wants to transmit '00'



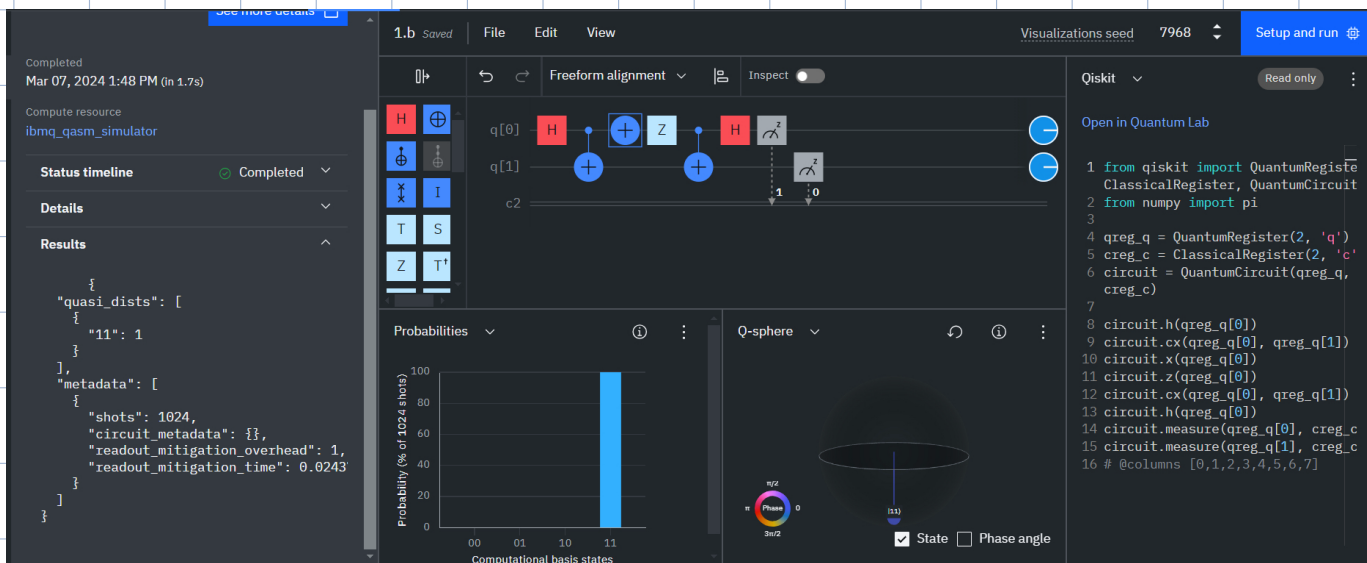
Alice wants to transmit '01'



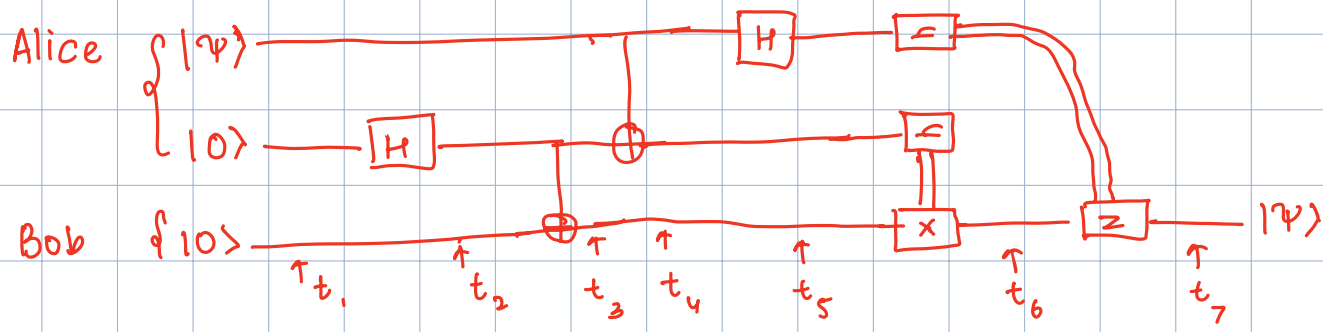
Alice transmits '0'



Alice transmits '1'



3.



$$|\psi\rangle = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle$$

$$t_1 = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle \quad (\otimes) |00\rangle$$

$$t_2 = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle \quad (\otimes) \frac{1}{\sqrt{2}} (|100\rangle + |110\rangle)$$

$$t_3 = \frac{\sqrt{3}}{2} |0\rangle + \frac{1}{2} |1\rangle \quad (\otimes) \frac{1}{\sqrt{2}} (|100\rangle + |111\rangle)$$

$$t_3 = \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} |1000\rangle + \frac{\sqrt{3}}{2} |1011\rangle + \frac{1}{2} |1100\rangle + \frac{1}{2} |1111\rangle \right)$$

$$t_4 = \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2} |1000\rangle + \frac{\sqrt{3}}{2} |1011\rangle + \frac{1}{2} |1110\rangle + \frac{1}{2} |1101\rangle \right)$$

$$t_5 = \frac{1}{\sqrt{2}} \left(\frac{\sqrt{3}}{2\sqrt{2}} (|10\rangle + |11\rangle) \quad (\otimes) |00\rangle + \frac{\sqrt{3}}{2\sqrt{2}} (|10\rangle + |11\rangle) \quad (\otimes) |11\rangle + \right.$$

$$\frac{1}{2\sqrt{2}} (|0\rangle - |1\rangle) \otimes |10\rangle + \frac{1}{2\sqrt{2}} (|0\rangle - |1\rangle) \otimes |01\rangle$$

$$t_5 = \frac{1}{4} \left(\sqrt{3} |000\rangle + \sqrt{3} |100\rangle + \sqrt{3} |011\rangle + \sqrt{3} |111\rangle + |010\rangle - |110\rangle + |001\rangle - |101\rangle \right)$$

For state t_6 and t_7 , Bob will do the calculation

$$t_7 = \frac{1}{4} (\sqrt{3} |0\rangle + \sqrt{3} |0\rangle + \sqrt{3} |0\rangle + \sqrt{3} |0\rangle + |1\rangle - |1\rangle + |1\rangle - |1\rangle)$$

$$= \frac{4}{4} (\sqrt{3} |0\rangle)$$

$$= \sqrt{3} |0\rangle$$

Generalized form

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$t_1 = |\psi\rangle \otimes |00\rangle$$

$$t_2 = |\psi\rangle \otimes \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$t_3 = |\psi\rangle \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$t_3 = (\alpha |0\rangle + \beta |1\rangle) \otimes \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

$$t_3 = \frac{1}{\sqrt{2}} (\alpha |000\rangle + \alpha |011\rangle + \beta |100\rangle + \beta |111\rangle)$$

$$t_4 = \frac{1}{\sqrt{2}} (\alpha |000\rangle + \alpha |011\rangle + \beta |110\rangle + \beta |101\rangle)$$

$$t_5 = \frac{1}{\sqrt{2}} \left\{ \frac{\alpha}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |00\rangle + \left(\frac{\alpha}{\sqrt{2}} (|0\rangle + |1\rangle) \otimes |11\rangle + \frac{\beta}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes |10\rangle + \frac{\beta}{\sqrt{2}} (|0\rangle - |1\rangle) \otimes |01\rangle \right) \right\}$$

$$t_5 = \frac{1}{2} (\alpha |000\rangle + \alpha |100\rangle + \alpha |011\rangle + \alpha |111\rangle + \beta |010\rangle - \beta |110\rangle + \beta |001\rangle - \beta |101\rangle)$$

$$t_7 = \frac{1}{2} (\alpha |0\rangle + \alpha |0\rangle + \alpha |0\rangle + \alpha |0\rangle + \beta |1\rangle - \beta |1\rangle + \beta |1\rangle - \beta |1\rangle)$$

$$= 2\alpha |0\rangle$$